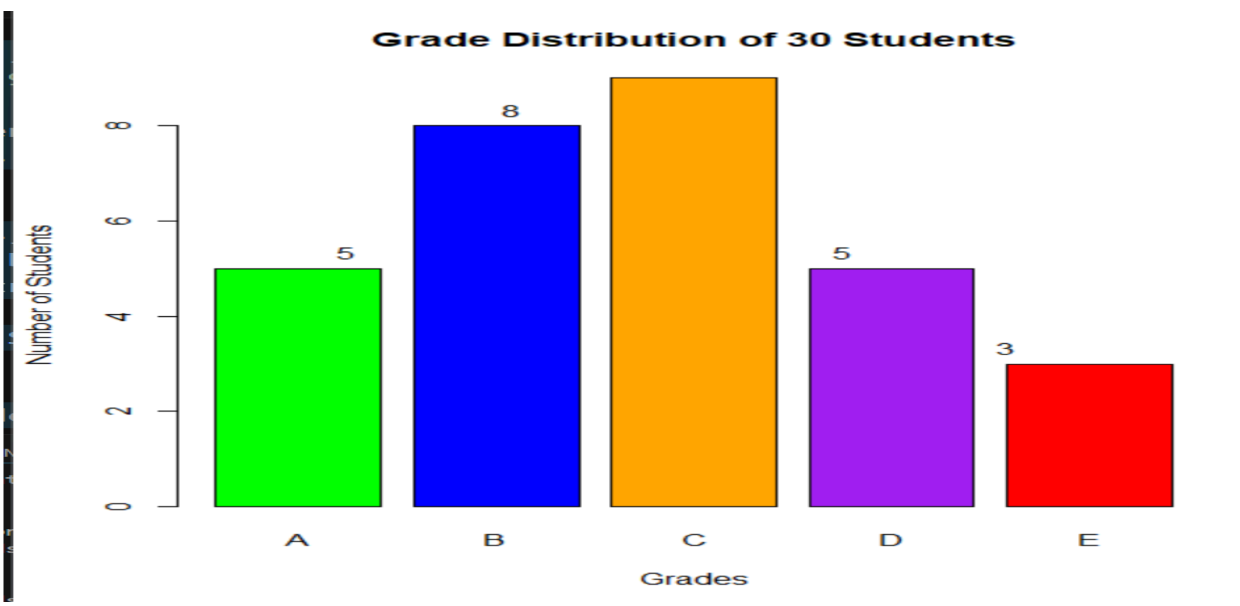
2.QUESTION

```
grades <- c("A", "B", "C", "D", "E")
students <- c(5, 8, 9, 5, 3)
marks <- c(95, 84.5, 74.5, 64.5, 50)
average <- sum(students * marks) / sum(students)
cat("Estimated Class Average:", round(average, 2), "\n")
barplot(
  students,
  names.arg = grades,
  col = c("green", "blue", "orange", "purple", "red"),
  main = "Grade Distribution of 30 Students",
  xlab = "Grades",
  ylab = "Number of Students"
)
text(
  x = seq_along(students),
  y = students,
  labels = students,
  pos = 3
)
```

OUTPUT:

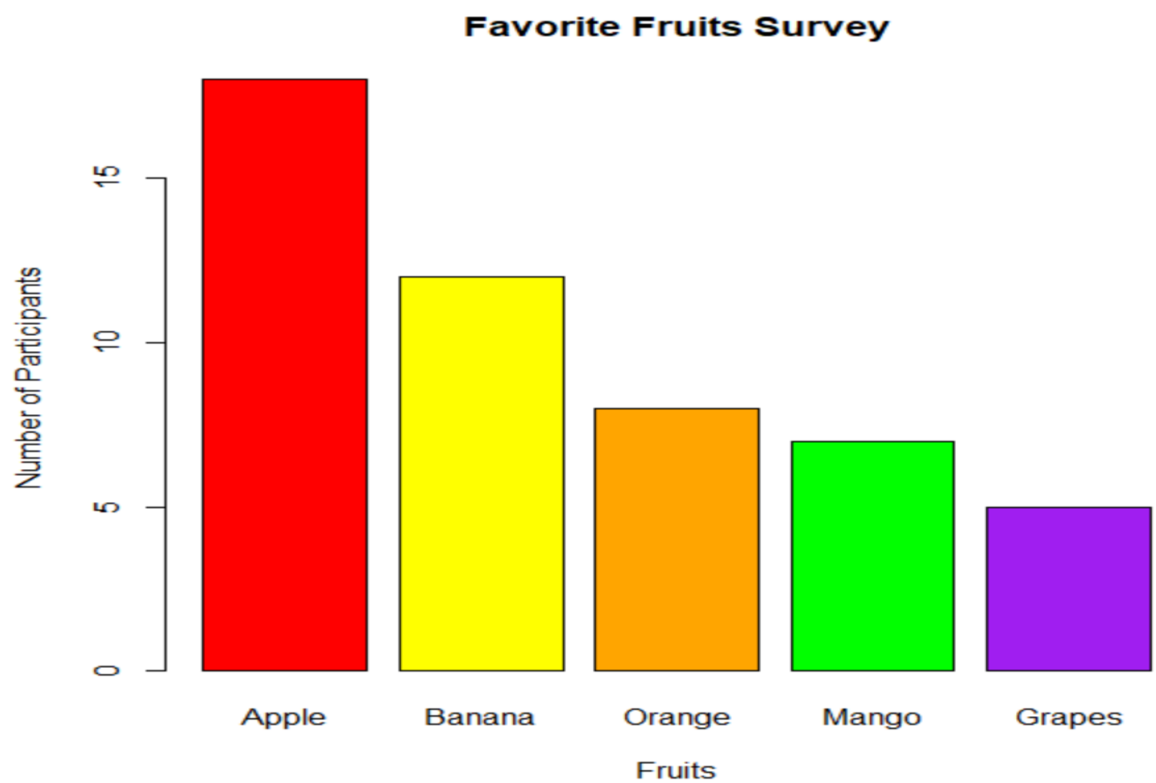


3.QUESTION

```r
fruits <- c("Apple", "Banana", "Orange", "Mango", "Grapes")
votes <- c(18, 12, 8, 7, 5)
barplot(votes,
        names.arg = fruits,
        col = c("red", "yellow", "orange", "green", "purple"),
        main = "Favorite Fruits Survey",
        xlab = "Fruits",
        ylab = "Number of Participants")
most_preferred <- fruits[which.max(votes)]
most_votes <- max(votes)
least_preferred <- fruits[which.min(votes)]
least_votes <- min(votes)
cat("Most Preferred Fruit :", most_preferred,
    "with", most_votes, "votes\n")

cat("Least Preferred Fruit :", least_preferred,
    "with", least_votes, "votes\n")
```

OUTPUT:



4.QUESTION

```
departments <- c("HR", "Finance", "Marketing", "IT", "Sales")
percent <- c(15, 20, 25, 30, 10)
total_employees <- 200
employees <- (percent / 100) * total_employees
result <- data.frame(
  Department = departments,
  Percentage = percent,
  Employees = employees
)
print(result)
marketing_employees <- employees[departments == "Marketing"]
cat("\nEmployees in Marketing Department =", marketing_employees)
```

OUTPUT:

```
  Department Percentage Employees
1         HR         15        30
2    Finance         20        40
3  Marketing         25        50
4         IT         30        60
5      Sales         10        20

> ployees in Marketing Department = 50
```

5.QUESTION

```r
activities <- c("Video Games", "Reading", "Cooking", "Painting", "Music")
percent <- c(20, 35, 20, 10, 15)
pie(percent,
    labels = paste(activities, percent, "%"),
    col = rainbow(5),
    main = "Employees' Weekend Free Time")
most_activity <- activities[which.max(percent)]
most_percent <- max(percent)
least_activity <- activities[which.min(percent)]
least_percent <- min(percent)
cat("Most Preferred Activity :", most_activity,
    "-", most_percent, "%\n")
cat("Least Preferred Activity :", least_activity,
    "-", least_percent, "%\n")
```

OUTPUT: